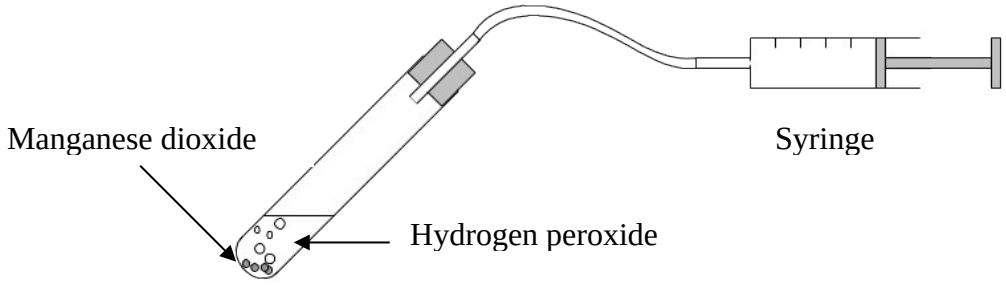
Rates of Reaction Questions

Method

Dilute hydrogen peroxide was put into a boiling tube and a small amount of manganese dioxide added. A delivery tube connected to a syringe was quickly fitted to the boiling tube and the time taken to get 10 mL of gas in the syringe was measured.



The temperature of the hydrogen peroxide was changed to see if the temperature of the reactants affected the rate of gas production.

Time (seconds)	**31**	**24**	**18**	**13**	**9**	**6**	**4**
Temperature (^{o}C)	**20**	**25**	**30**	**35**	**40**	**45**	**50**

1. Name the gas produced in the reaction.

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2. Write a conclusion for the experiment.

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3. The **<u>Method</u>** describing how the experiment was done (at the beginning of Part B) failed to ensure that it was either a **valid** or **reliable** experiment.

(a) What should have been done to make the results valid?

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(b) What should have been done to make the results reliable?

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4. A second investigation was done on this reaction, but this time the amount of MnO_2 used was varied while all other factors were kept constant. The student's result suggested that the amount of MnO_2 added made no difference to the rate of decomposition of H_2O_2.

(a) Assuming that the student's results in both experiments were valid and reliable, what can you say about the role of the MnO_2 in the reaction?

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(b) Suggest another way that you could increase the rate of decomposition (apart from what ` has already been done in the two experiments).

_____

_____Part

C: Hydrogen.

Draw a labelled diagram of the apparatus you would set up to collect a test tube of hydrogen gas. The technique you are to use is called "downward displacement of water." Clearly label the chemicals used.